

Exercise 6-1: Clean the Polling data

Did you use a language model?

A : No, I did not find any use for a language model for this assignment except for writing comments.

Read the data

```
In [1]: # Imports the Pandas library  
import pandas as pd
```

```
In [2]: # Load the polling data for the 2016 presidential election from a CSV file into  
polls = pd.read_csv('president_polls_2016.csv')
```

Examine the data

```
In [3]: # Display a concise summary of the DataFrame, including the number of non-null e  
polls.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12624 entries, 0 to 12623
Data columns (total 27 columns):
#   Column                Non-Null Count  Dtype
---  -
0   cycle                  12624 non-null  int64
1   branch                 12624 non-null  object
2   type                   12624 non-null  object
3   matchup                12624 non-null  object
4   forecastdate           12624 non-null  object
5   state                  12624 non-null  object
6   startdate              12624 non-null  object
7   enddate                12624 non-null  object
8   pollster               12624 non-null  object
9   grade                  11337 non-null  object
10  samplesize             12621 non-null  float64
11  population              12624 non-null  object
12  poll_wt                 12624 non-null  float64
13  rawpoll_clinton         12624 non-null  float64
14  rawpoll_trump           12624 non-null  float64
15  rawpoll_johnson         8397 non-null   float64
16  rawpoll_mcmullin        90 non-null     float64
17  adjpoll_clinton         12624 non-null  float64
18  adjpoll_trump           12624 non-null  float64
19  adjpoll_johnson         8397 non-null   float64
20  adjpoll_mcmullin        90 non-null     float64
21  multiversions           36 non-null     object
22  url                     12621 non-null  object
23  poll_id                 12624 non-null  int64
24  question_id             12624 non-null  int64
25  createddate             12624 non-null  object
26  timestamp               12624 non-null  object
dtypes: float64(10), int64(3), object(14)
memory usage: 2.6+ MB

```

```

In [4]: # Display the number of unique entries in each column of the DataFrame.
        polls.nunique()

```

```
Out[4]: cycle          1
        branch         1
        type           3
        matchup        1
        forecastdate    1
        state          57
        startdate      352
        enddate        345
        pollster       196
        grade          10
        samplesize     1766
        population      4
        poll_wt        4399
        rawpoll_clinton 1312
        rawpoll_trump   1385
        rawpoll_johnson 584
        rawpoll_mcmullin 16
        adjpoll_clinton 12569
        adjpoll_trump   12582
        adjpoll_johnson 6629
        adjpoll_mcmullin 57
        multiversions   1
        url            1304
        poll_id        4208
        question_id     4208
        createddate     222
        timestamp       3
        dtype: int64
```

Drop columns and rows

```
In [5]: # Filter the DataFrame to include only rows where the type of poll is 'now-cast'
        polls = polls.query('type == "now-cast"')
```

```
In [6]: # Display the number of unique entries in each column of the filtered DataFrame.
        polls.nunique()
```

```
Out[6]: cycle          1
        branch         1
        type           1
        matchup        1
        forecastdate    1
        state          57
        startdate      352
        enddate        345
        pollster       196
        grade          10
        samplesize     1766
        population      4
        poll_wt        3894
        rawpoll_clinton 1312
        rawpoll_trump   1385
        rawpoll_johnson 584
        rawpoll_mcmullin 16
        adjpoll_clinton 4202
        adjpoll_trump   4199
        adjpoll_johnson 2210
        adjpoll_mcmullin 30
        multiversions   1
        url            1304
        poll_id        4208
        question_id     4208
        createddate     222
        timestamp       1
        dtype: int64
```

```
In [7]: # Remove specific columns that are not needed for further analysis from the Data
        polls.drop(columns=['cycle', 'branch', 'type', 'matchup',
                           'forecastdate', 'multiversions', 'timestamp'], inplace=True)
```

```
In [8]: # Filter the DataFrame to exclude rows where the state is 'U.S.', focusing on st
        polls = polls.query('state != "U.S."')
```

```
In [9]: # Display the number of unique entries in each column of the DataFrame after fil
        polls.nunique()
```

```
Out[9]: state          56
startdate          245
enddate            241
pollster           176
grade              10
samplesize        1308
population          4
poll_wt            2874
rawpoll_clinton    1162
rawpoll_trump       1217
rawpoll_johnson     578
rawpoll_mcmullin    16
adjpoll_clinton     3100
adjpoll_trump       3097
adjpoll_johnson     1672
adjpoll_mcmullin    30
url                847
poll_id            3102
question_id        3102
createddate        185
dtype: int64
```

Rename columns

```
In [10]: # Rename specific columns related to raw polling numbers to more descriptive names
polls_names_dict = {'rawpoll_clinton': 'clinton_pct',
                    'rawpoll_trump': 'trump_pct',
                    'rawpoll_johnson': 'johnson_pct',
                    'rawpoll_mcmullin': 'mcmullin_pct'}
polls.rename(columns=polls_names_dict, inplace=True)
```

Fix data types

```
In [11]: # Display the DataFrame's concise summary again to confirm the column names have
polls.info()
```

```

<class 'pandas.core.frame.DataFrame'>
Index: 3102 entries, 4216 to 8415
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
0   state                  3102 non-null   object
1   startdate              3102 non-null   object
2   enddate                3102 non-null   object
3   pollster               3102 non-null   object
4   grade                  2961 non-null   object
5   samplesize             3101 non-null   float64
6   population             3102 non-null   object
7   poll_wt                3102 non-null   float64
8   clinton_pct            3102 non-null   float64
9   trump_pct              3102 non-null   float64
10  johnson_pct            2246 non-null   float64
11  mcmullin_pct           30 non-null     float64
12  adjpoll_clinton        3102 non-null   float64
13  adjpoll_trump          3102 non-null   float64
14  adjpoll_johnson        2246 non-null   float64
15  adjpoll_mcmullin       30 non-null     float64
16  url                    3102 non-null   object
17  poll_id                3102 non-null   int64
18  question_id            3102 non-null   int64
19  createddate            3102 non-null   object
dtypes: float64(10), int64(2), object(8)
memory usage: 508.9+ KB

```

```

In [12]: # Create a list of column names that represent dates.
date_cols = ['startdate', 'enddate', 'createddate']

```

```

In [14]: # Convert the columns in 'date_cols' list to datetime objects with a specified f
polls[date_cols] = polls[date_cols].apply(lambda x: pd.to_datetime(x, format='%Y

```

```

In [15]: # Convert the 'state' column to a categorical type to optimize memory usage and
polls.state = polls.state.astype('category')

```

```

In [16]: # Convert the 'population' column to a categorical type for the same reasons as
polls.population = polls.population.astype('category')

```

```

In [17]: # Display the DataFrame's summary again to confirm the data type changes.
polls.info()

```

```

<class 'pandas.core.frame.DataFrame'>
Index: 3102 entries, 4216 to 8415
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
0   state                  3102 non-null   category
1   startdate              3102 non-null   datetime64[ns]
2   enddate                3102 non-null   datetime64[ns]
3   pollster               3102 non-null   object
4   grade                  2961 non-null   object
5   samplesize             3101 non-null   float64
6   population             3102 non-null   category
7   poll_wt                3102 non-null   float64
8   clinton_pct            3102 non-null   float64
9   trump_pct              3102 non-null   float64
10  johnson_pct            2246 non-null   float64
11  mcmullin_pct           30 non-null     float64
12  adjpoll_clinton        3102 non-null   float64
13  adjpoll_trump          3102 non-null   float64
14  adjpoll_johnson        2246 non-null   float64
15  adjpoll_mcmullin       30 non-null     float64
16  url                    3102 non-null   object
17  poll_id                3102 non-null   int64
18  question_id            3102 non-null   int64
19  createddate            3102 non-null   datetime64[ns]
dtypes: category(2), datetime64[ns](3), float64(10), int64(2), object(3)
memory usage: 469.2+ KB

```

In [18]: polls

Out[18]:

	state	startdate	enddate	pollster	grade	samplesize	population	poll_v
4216	New Mexico	2016-11-06	2016-11-06	Zia Poll	NaN	8439.0	lv	4.60949
4221	Virginia	2016-11-03	2016-11-04	Public Policy Polling	B+	1238.0	lv	3.92352
4223	Iowa	2016-11-01	2016-11-04	Selzer & Company	A+	800.0	lv	3.84223
4225	Wisconsin	2016-10-26	2016-10-31	Marquette University	A	1255.0	lv	3.78995
4226	North Carolina	2016-11-04	2016-11-06	Siena College	A	800.0	lv	3.77413
...	...	...	...	...	...	...	...	...
8411	North Carolina	2016-08-26	2016-08-27	Public Policy Polling	B+	1177.0	lv	0.00000
8412	North Carolina	2016-03-09	2016-03-10	SurveyUSA	A	1576.0	lv	0.00000
8413	North Carolina	2016-07-31	2016-08-02	SurveyUSA	A	400.0	lv	0.00000
8414	Utah	2016-08-05	2016-08-25	Ipsos	A-	177.0	lv	0.00000
8415	Missouri	2016-10-21	2016-11-02	Ipsos	A-	405.0	lv	0.00000

3102 rows × 20 columns

In []: